

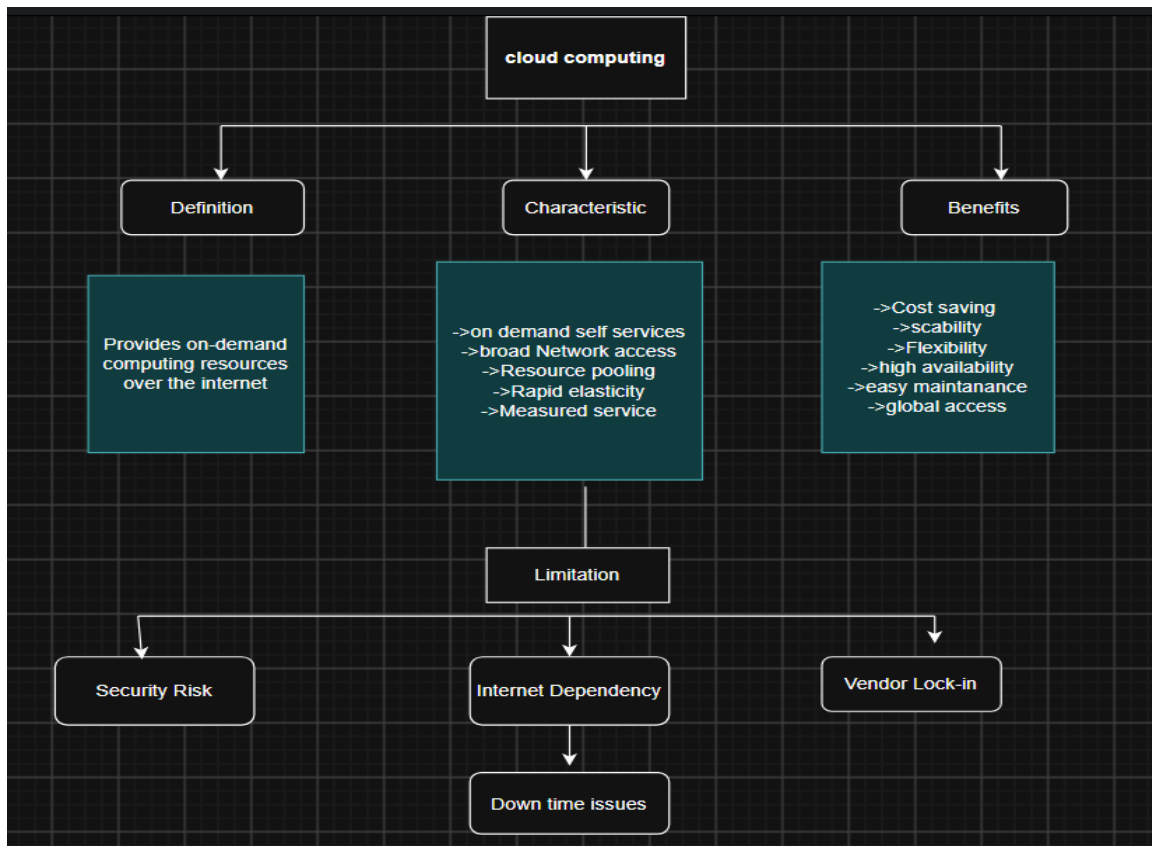
Course Name: Cloud Computing and Big Data Analytics

Course Code: CSA1504

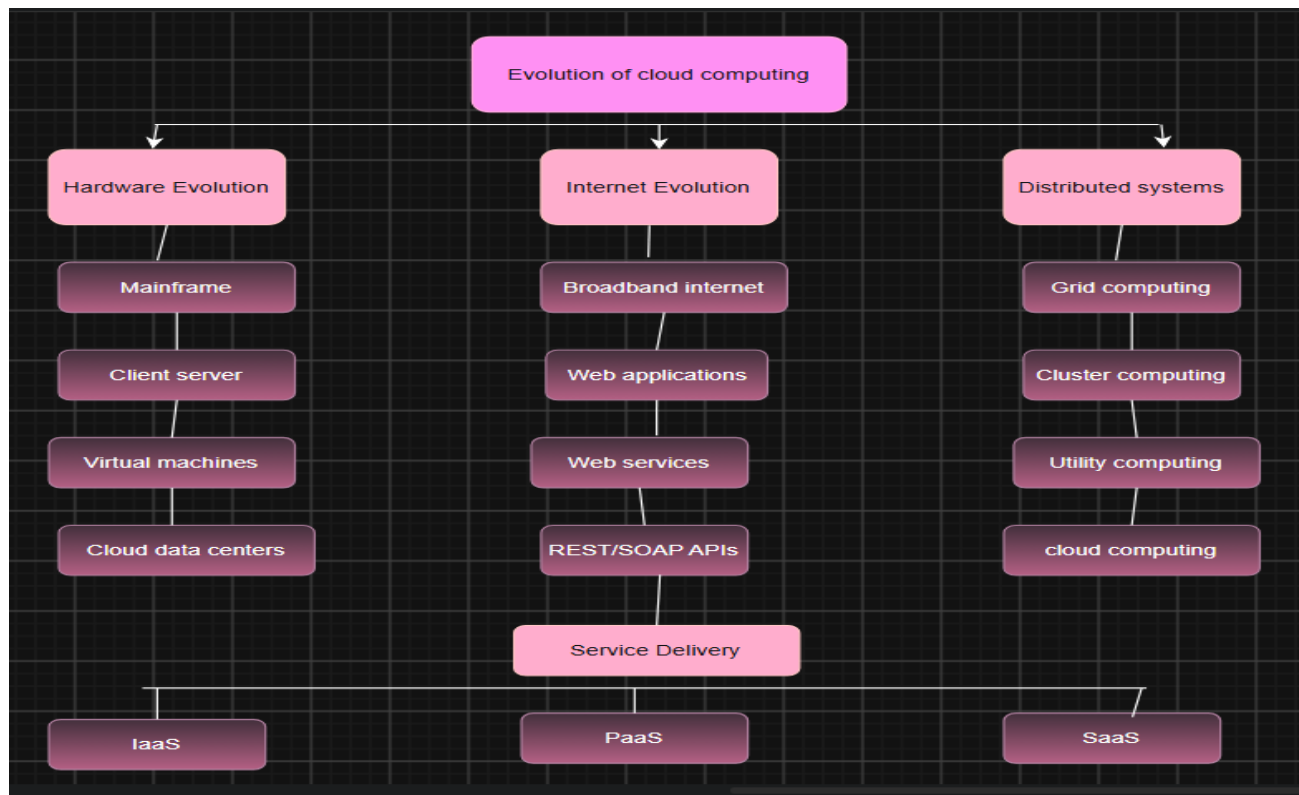
Assessment: Assessment Tool 1: Concept Mapping

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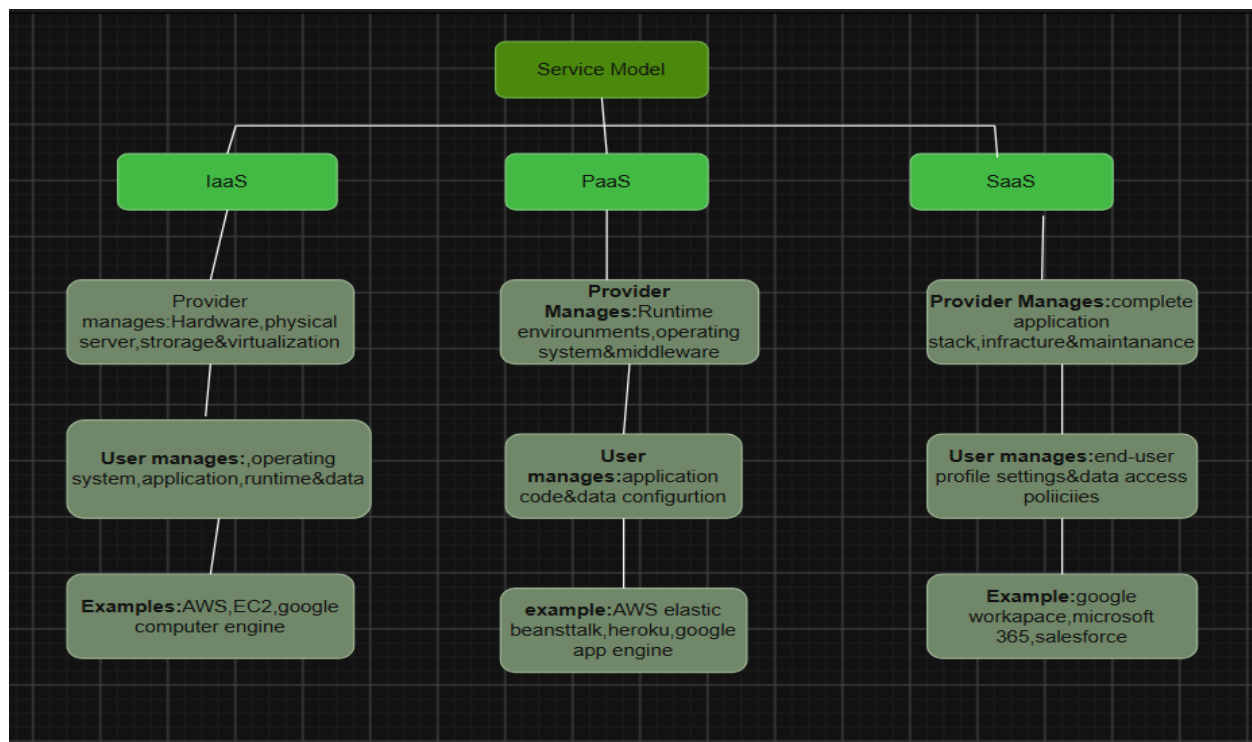
1.Cloud Computing Fundamentals.



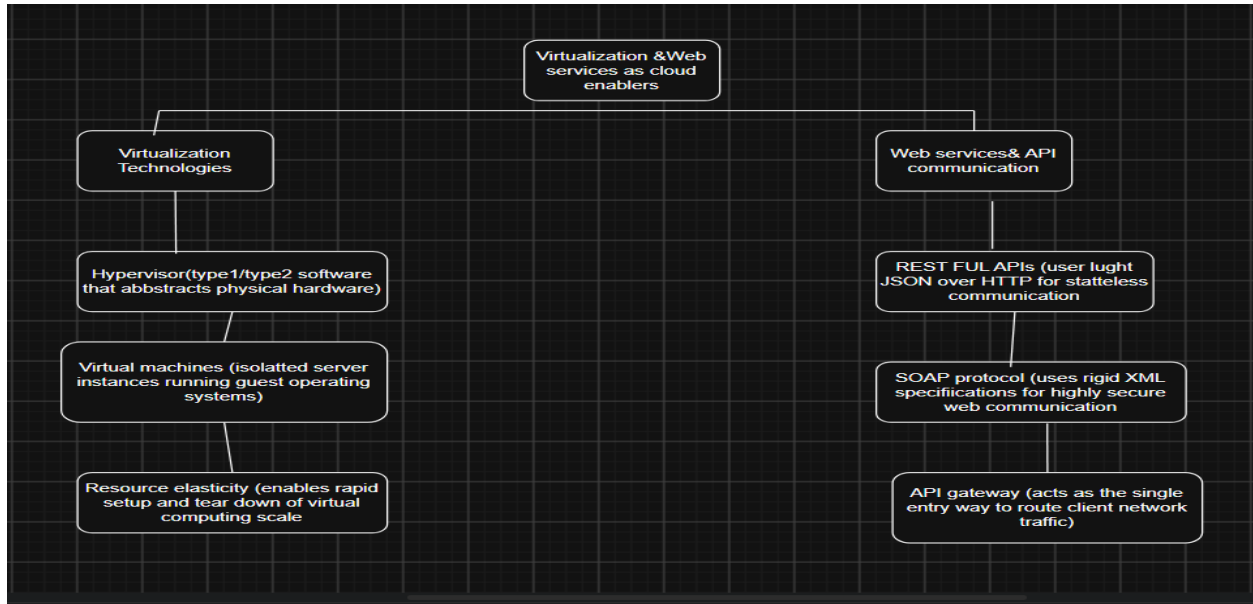
2.Evolution of Cloud Computing.



3: Service Models.



4.Virtualization and Web Services as Enablers of Cloud.



Task 2A: Service Model Responsibility Matrix

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Cloud Service	Managed by Cloud Provider	Managed by Student Team	Why Suitable for the Project
SaaS (Software as a Service)	Provides and maintains software applications, updates, and security	Uses the software for project activities	Enables quick communication and collaboration
PaaS (Platform as a Service)	Manages operating system, runtime environment, middleware, and platform	Develops, tests, and deploys application code	Simplifies application development and deployment
IaaS (Infrastructure as a Service)	Provides virtual machines, storage, networking, and servers	Configures virtual machines, installs software, and manages applications	Offers complete control over infrastructure
FaaS (Function as a Service / Serverless)	Executes serverless functions, scales automatically, and manages infrastructure	Writes and deploys functions for project logic	Efficiently processes attendance events without server management
Database as a Service (DBaaS)	Maintains database servers, backups, availability, and security	Creates tables, manages queries, and stores project data	Securely stores attendance records and student information
Storage as a Service (STaaS)	Provides cloud storage, file management, replication, and backup	Uploads reports, images, and project documents	Ensures reliable and scalable file storage
Identity as a Service (IDaaS)	Handles authentication, authorization, and user identity management	Defines user roles, permissions, and access policies	Provides secure login for students, faculty, and administrators
Monitoring & Logging Service	Collects performance metrics, logs, alerts, and system monitoring	Reviews logs, identifies issues, and improves application performance	Helps monitor system health and troubleshoot problems efficiently

Cloud Service Decision Matrix

Project Module	Cloud Service	Example Service	Reason
User Authentication	IDaaS	Firebase Authentication	Secure login
Frontend Hosting	PaaS	Azure App Service / Netlify	Easy web hosting
Backend/API	PaaS	Azure App Service	Runs REST APIs
Database	DBaaS	Firebase Firestore / Azure SQL	Stores attendance data
Object Storage	Storage as a Service	Azure Blob Storage	Stores reports and files
Analytics	PaaS + AI	Azure Machine Learning	Predict attendance trends
Notification	SaaS	Firebase Cloud Messaging	Sends alerts to students
AI/Python Module	PaaS	Azure Machine Learning	Time-series prediction
Security	IDaaS	Firebase Authentication	Secure access
CI/CD	DevOps Service	GitHub Actions	Automatic deployment
Backup	Storage Service	Azure Backup	Data recovery
Monitoring	Monitoring Service	Azure Monitor	Track application health

Technical justification:

The architecture diagram and matrices demonstrate **CO1** by showing how cloud computing concepts are applied to design a secure, scalable, and efficient system. The selected **service model** is **Software as a Service (SaaS)**, where users access the application through a web interface without managing the underlying infrastructure. **APIs** enable communication between the user interface, application services, databases, and external cloud services, ensuring seamless integration. The system supports **elasticity** by automatically allocating additional resources during periods of high demand and releasing them when demand decreases. **Scalability** is achieved by distributing workloads across multiple virtual servers, allowing the application to support an increasing number of users. **Virtualization/managed compute services** improve resource utilization by hosting applications on virtual machines or cloud-managed platforms, reducing operational overhead. Finally, the **shared responsibility model** defines cloud responsibility boundaries: the cloud provider manages the physical infrastructure, networking, storage, and virtualization, while the customer is responsible for application configuration, user access, and data security. Together, these elements justify that the proposed cloud solution satisfies the intended learning outcome (CO1) by applying core cloud computing principles effectively.